PROOF TECHNIQUES

CSE 2213 – Discrete Mathematics

PROOF TECHNIQUES

- Three basic techniques –
  - Direct proof
  - Proof by contradiction
  - Proof by contraposition

DIRECT PROOF

- To prove $p \rightarrow q$, we first assume that p is true, and hence prove that q is true
  - We reach the conclusion directly from our assumption

EXAMPLE

- Give a direct proof of the theorem "If *n* is an even integer, then n^2 is even.”

EXAMPLE

- Give a direct proof that if m and n are both perfect squares, then nm is also a perfect square.(A perfect square is the square of an integer)

EXAMPLE
ROSEN, PAGE 85, EXERCISE 10

Prove that the product of two rational numbers is rational

- Note: The real number r is *rational* if there exist integers p and q with $q \neq 0$ such that $r = p / q$. A real number that is not rational is called *irrational.*

PROOF BY CONTRAPOSITION

- To prove $p \rightarrow q$, we first assume that $\neg q$ is true, and hence prove that $\neg p$ is true
  - We actually prove the contrapositive of the actual sentence, i.e. $\neg q \rightarrow \neg p$

EXAMPLE
ROSEN, PAGE 78, EXAMPLE 3

Prove that, for all integer n, if $3n + 2$ is odd, then n is odd

EXAMPLE

- Prove that if *n* is an integer and n^2 is odd, then *n* is odd.
- P: n^2 is odd
- Q:n is odd
- ~P: n^2 is even
- ~Q: n is even

PROOF BY CONTRADICTION

- For statements of the form "if p, then q", we assume p is true and $\neg q$ is true (that is, q is false)
- Then, assuming that $\neg q$ is true, we show that p cannot be true
  - This will be a contradiction to our assumption that p is true
- Therefore, our assumption that when p is true then $\neg q$ is true, is false
- Therefore, when p is true, q must also be true

PROOF BY CONTRADICTION

- Give a proof by contradiction of the theorem "If 3n + 2 is odd, then *n* is odd."
- We assume that 3n+2 is odd and n is even

EXAMPLE

- Give a proof by contradiction of the theorem "If 3n + 2 is even, then *n* is even."

VACUOUS AND TRIVIAL PROOFS

- If we prove a conditional statement by disproving its hypothesis, then the proof technique is called vacuous proof
- If we prove a conditional statement by proving its conclusion, then the proof technique is called trivial proof

QUICK EXERCISE

- Is $P(0)$ true, where $P(n) \equiv (n > 1) \to (n^2 > n)$?
  - Vacuous proof
- Is $P(0)$ true, where $P(n) \equiv \forall a \forall b \big((a \geq b) \to (a^n \geq b^n)\big)$?
  - Trivial proof

DISPROVING UNIVERSALLY QUANTIFIED PROPOSITIONS

- Prove that "Every positive integer is the sum of the squares of two integers" is false.
  - We just find a counterexample, that invalidates the use of "every" in the sentence
  - Here, a counterexample is 3 (because 3 = 1 + 2, and 2 is not the square of any integer)

EXAMPLE
ROSEN, PAGE 85, EXERCISE 5

Prove that, if $m + n$ and $n + p$ are even integers, then $m + p$ is also even

EXAMPLE

- Proof by contradiction: "if mn even, then m even or n even".

EXAMPLE
ROSEN, PAGE 85, EXERCISE 13

Prove that, if x is irrational, then $\frac{1}{x}$ is also irrational.
(Given that 1/x will never be equal to 0)

EXAMPLE
ROSEN, PAGE 85, EXERCISE 9

Prove that the sum of an irrational and a rational number is irrational